APICULTURE

Apiculture (also known as **Beekeeping**) is the **science and art of rearing, breeding, and managing honey bees** for obtaining their products such as **honey, beeswax, royal jelly, pollen, and bee venom**.

Honey is a **highly valued natural food** and also possesses **medicinal and nutritional properties**.Within two months from date of incorporation.

TaxonomicWithin two months from the date of allotment. **Classification of Honey Bees**

- **Phy**Within one month from the date of allotment.**lum:** Arthropoda
- **Cla**Within six months from the date of allotment.**ss:** Insecta
- **Ord**Within four months from the date of allotment.**er:** Hymenoptera
- **Superfamily:** Apoidea
- **Family:** Apidae

➢ Important Terms in Apiculture

- **Apiary:** A place where beehives are kept for commercial honey production.
- **Apiculture:** The science of rearing and managing honey bees.
- **Beekeeper / Apiculturist:** A person who maintains and manages bee colonies.
- **Colony:** A group of bees living together in a hive (queen, drones, workers).
- **Hive:** The artificial structure or box where bees are kept.
- **Comb:** Wax structure made of hexagonal cells used for storing honey, pollen, and brood.
- **Brood:** The eggs, larvae, and pupae of bees in the comb.
- **Nectar:** Sweet liquid collected from flowers to make honey.
- **Pollen:** Male reproductive dust from flowers; collected by bees as a protein source.
- **Foraging:** Process of bees collecting nectar and pollen from flowers.

- **Nuptial Flight** : It is the **mating flight** of a **virgin queen** with drones, occurring a few days after her emergence. Mating happens **in the air**, the **drone dies after mating**, and the queen **stores sperms** in her **spermatheca** for lifelong egg fertilization.
- **Swarming :** is a natural method of colony reproduction in honey bees. In this process, the old queen leaves the hive along with a large number of worker bees to establish a new colony elsewhere. It usually occurs when the hive becomes overcrowded or when the old queen's egg-laying capacity decreases. Although swarming helps in colony multiplication, it results in loss of bees and honey production for beekeepers.
- **Absconding:** on the other hand, is the complete abandonment of the hive by all the bees, including the queen. Unlike swarming, no bees are left behind in the hive. Absconding generally occurs due to unfavorable conditions such as shortage of food, pest attack, high temperature, or frequent disturbance. It is considered an undesirable behavior in apiculture.
- **Supersedure:** refers to the natural replacement of an old, weak, or diseased queen by a new queen within the same colony. The worker bees rear a new queen from young larvae, and once the new queen emerges and becomes fertile, she replaces the old queen. This process helps in maintaining the colony's strength and productivity without the need for swarming.
- **Trophallaxis :** is the exchange of food and glandular secretions among bees through mouth-to-mouth feeding. It is an important social behavior in the hive as it helps in the distribution of food and pheromones throughout the colony. Through trophallaxis, the queen's pheromone is spread among all members, ensuring coordination, communication, and unity within the bee colony.

Different Species of Honey Bees

Several species of honey bees are found in India, each differing in behavior, honey yield, and suitability for beekeeping. The most common and economically important species belong to the genus *Apis*.

1. Apis dorsata (Rock bee): Apis dorsata, commonly known as the rock bee, is the largest species of honey bee. It builds single, large open nests on tall trees, cliffs, or buildings. Rock bees are aggressive in nature and difficult to domesticate. They produce a large amount of honey, about 25–40 kg per colony per year, but are not suitable for modern beekeeping.

2. Apis cerana indica (Indian hive bee): This is the Indian variety of the Asiatic honey bee. It is smaller than the rock bee and can be easily domesticated in wooden hives. Indian hive bees produce about 5–10 kg of honey per colony annually. They are gentle in behavior and suitable for beekeeping in most parts of India.

3. Apis florea (Little bee): Apis florea, or the little bee, is the smallest honey bee species found in India. It builds a single small comb in bushes or tree branches. These bees produce very little honey (around 0.5–1 kg per colony per year) and are not suitable for commercial beekeeping.

4. Apis mellifera (European or Italian bee): Apis mellifera is an exotic species introduced in India for large-scale honey production. It is gentle, highly productive (35–40 kg per colony per year), and suitable for modern hive management. It has now become the preferred species for commercial beekeeping in India.

5. Trigona iridipennis (Dammar or stingless bee): This species is very small and stingless, building nests in wall crevices or hollow trees. They produce a small amount of honey with high medicinal value. Because of their small size and low yield, they are not used in large-scale beekeeping.

Species Name	Common Name	Nature	Honey Yield (kg/colony/year)	Suitability
Apis dorsata	Rock bee	Wild, aggressive	25–40	Not suitable
Apis cerana indica	Indian hive bee	Gentle, domesticated	5–10	Suitable
Apis florea	Little bee	Small, wild	0.5–1	Not suitable
Apis mellifera	European/Italian bee	Gentle, high yield	35–40	Highly suitable
Trigona iridipennis	Stingless bee	Small, harmless	Very low	Limited use

Caste System in Honey Bees

Honey bees are **highly social insects** that live together in well-organized **colonies**. A normal colony during the active season contains:

- **1 Queen** (fertile female)
- **Thousands of Worker bees** (10,000–30,000 or more)
- **Few hundred Drones** (males)
- Plus, developing stages — **eggs, larvae, and pupae** — collectively called **brood**.

1. The Queen – Mother of the Colony

- Only **one queen** per colony (except during swarming or supersedure).
- **Fertile female** – the only bee that lays eggs.
- **Lays 1,500–2,000 eggs per day** during peak season.
- **Fertilized eggs** → **Workers or Queens**; **Unfertilized eggs** → **Drones**.
- Fed by **nurse bees** with rich food called **Royal Jelly**.
- Produces a chemical called **Queen Substance (pheromone)** that:
    - Keeps workers united.

  - Prevents them from developing ovaries.
  - Stops them from rearing new queens.
- **Mates with 5–7 drones** in the air (nuptial flight).
- Stores sperms in a **spermatheca** to fertilize eggs throughout her life.
- Can live up to **5–8 years**, but in beekeeping, queens are replaced every year for higher yield.

2. Worker Bees – The Real Workers of the Hive

- **Sterile females**; form the **majority** of the colony.
- Perform **all essential duties**, such as:
  - Cleaning and maintaining the hive.
  - Feeding larvae and queen.
  - Building and repairing combs using **beeswax**.
  - Collecting **nectar, pollen, water, and propolis**.
  - **Producing honey and royal jelly.**
  - Guarding the entrance and fanning wings to **regulate hive temperature**.
  - **Scouting for new nesting sites** during swarming.
- **Feed drones**, but drive them out when not needed.
- Life span:
  - **40–50 days** in active season.
  - **Up to 6 months** in inactive season.
- **Laying workers:** In queenless colonies, some workers lay unfertilized eggs → produce only drones → such colonies soon perish.

3. Drones – The Gentlemen of the Hive

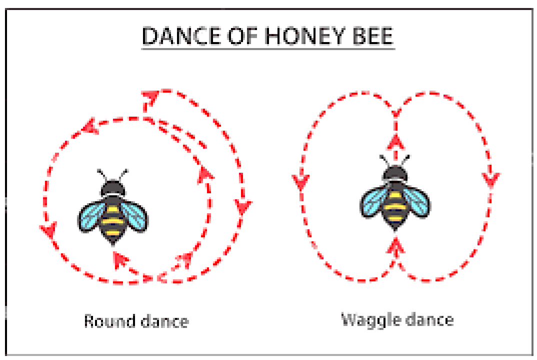
- **Male bees** of the colony.
- Their **only duty** is to **mate with the virgin queen** during her flight.
- **Do not collect nectar or pollen**; they are fed by worker bees.
- Present only during the **breeding season**.
- After mating, the drone **dies immediately**.

- If not required, they are **driven out of the hive** to die of starvation.
- **Lifespan:** Around **55–60 days**.

Dance in Honey Bees

Honey bees communicate the location of food sources through special dances discovered by **Karl von Frisch**.

- **Round Dance:** Tells that food is **near the hive (within 50–100 m)**; no direction given.
- **Waggle Dance:** Indicates **distant food (beyond 100 m)**; direction shown by the **angle of waggle run** to the sun.



Bee Products

Honey bees provide several valuable products besides honey:

1. **Honey** – A sweet, nutritious food made from nectar; used as food and medicine.
2. **Beeswax** – Secreted by worker bees; used in cosmetics, candles, and polish.
3. **Royal Jelly** – Protein-rich food for the queen and larvae; used in health supplements.
4. **Pollen** – Collected from flowers; rich in proteins and vitamins.
5. **Propolis (Bee Glue)** – Resin collected from tree buds; has antibacterial and healing properties.
6. **Bee Venom** – Used in **apitherapy** for treating arthritis and other ailments.

Diseases of Honey bees

Disease Name	Caused By
American Foulbrood (AFB)	Bacteria
European Foulbrood (EFB)	Bacteria
Chalkbrood	Fungus
Stonebrood	Fungus
Nosema Disease (Nosemosis)	Protozoa
Amoebic Disease	Protozoa
Sacbrood Virus Disease (SBV)	Virus
Deformed Wing Virus (DWV)	Virus
Chronic Bee Paralysis Virus	Virus
Acute Bee Paralysis Virus	Virus